Professional Installation Manual

Ubuntu Server Installation Manual

A complete guide for deploying Ubuntu Server 26.04 LTS in a virtual machine

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ITEP 414, System Administration and Maintenance

Week 3 Portfolio Project, Mini Project

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1. Introduction

1.1 Purpose of this manual

This manual describes how to deploy a clean Ubuntu Server 26.04 LTS installation inside Oracle VirtualBox. It is written so that any system administrator can follow the steps without additional instructions, even without prior experience with Ubuntu. The manual covers the full lifecycle of the initial deployment: creating the virtual machine, installing the operating system, configuring the server, verifying that it works, and resolving the most common problems.

1.2 Audience

This manual is intended for junior system administrators, IT support engineers, and students who need to stand up a Linux server quickly and correctly. Basic familiarity with virtual machines and the Linux command line is helpful but not required.

1.3 Prerequisites and software requirements

Software	Version	Purpose
Oracle VirtualBox	7.x or newer	Runs the virtual machine
Ubuntu Server ISO	26.04 LTS	Operating system installer
Host machine	Windows, macOS, or Linux	8 GB RAM recommended

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2. Ubuntu Server Installation Guide

2.1 Creating the virtual machine

1. Download the Ubuntu Server LTS ISO from https://ubuntu.com/download/server.
2. Open VirtualBox and click **New**.
3. Enter the machine name, for example Ubuntu-Server-Week03, select **Linux** as the type and **Ubuntu (64-bit)** as the version.
4. Allocate **4096 MB** of base memory and **2** virtual CPUs.
5. Create a virtual hard disk of **40 GB** using the VDI format.
6. Open **Settings**, go to **Storage**, and attach the downloaded ISO to the optical drive.
7. Confirm the network adapter is set to **NAT**, then click **Start**.

Allocate only the resources the server actually needs. The virtual machine takes memory and CPU away from the host while it is running.

2.2 Installing the operating system

1. From the boot menu, choose **Try or Install Ubuntu Server** and press Enter.
2. Select **English** as the language.

3. Select **English (US)** as the keyboard layout.
4. On the network screen, keep the default DHCP settings and **write down the IP address** that is assigned, because the same address is used later for SSH.
5. Enter the hostname, for example server01.
6. Create the administrator account with a real name, a username (for example the administrator's last name), and a strong password.
7. For disk setup, choose **Guided, use entire disk** and confirm the proposed layout.
8. On the SSH screen, enable **Install OpenSSH server** if remote administration is planned.
9. Skip the optional snap packages unless the server requires them.
10. Wait for the installation to finish.

The installation takes about 10 to 15 minutes. The installer writes the packages to disk at the end, so do not power off the machine during this phase.

2.3 Completing the installation

1. When the installer shows **Installation complete**, click **Reboot Now**.
2. When asked to remove the installation medium, confirm. If the machine boots into the installer again, remove the ISO from **Devices, Optical Drives** in the VirtualBox menu and reboot.
3. Wait for the first boot to finish. The system generates the SSH host keys automatically and then shows the login prompt server01 login:.

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3. Initial Server Configuration

3.1 First login and changing the password

Log in with the username and password created during installation. The prompt changes to user@server01:~$, which means the login succeeded. If the password was ever visible on screen or shared with someone, change it immediately:

```
passwd
```

The command asks for the current password and then the new one. Use a password of at least 12 characters with uppercase letters, lowercase letters, numbers, and symbols.

3.2 Enabling and securing SSH

If OpenSSH was enabled during installation, verify that the service is running and starts on boot:

```
sudo systemctl enable --now ssh
systemctl status ssh
```

If the service is not installed, install it first:

```
sudo apt install openssh-server -y
sudo systemctl enable --now ssh
```

The output of systemctl status ssh must show **active (running)** and **enabled**. When remote administration is no longer needed, disable the service with sudo systemctl disable --now ssh.

3.3 Updating the system

Apply all available security and software updates immediately after installation:

```
sudo apt update
sudo apt upgrade -y
```

Run these two commands regularly, ideally weekly, to keep the server protected. The output must end without errors and return to the prompt.

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4. Server Verification Procedures

Run the checks below after installation and record the results in the deployment notes. Each check has a command and the expected output.

4.1 Hostname check

```
hostname
```

Expected output: the configured hostname, for example server01.

4.2 Network check

```
ip addr
```

Expected output: an inet line under the network adapter (for example enp0s3) showing the assigned IPv4 address, such as 10.0.2.15.

4.3 Internet connectivity check

```
ping -c 4 google.com
```

Expected output: four reply lines and a summary showing **0% packet loss**. Any loss means the network configuration must be reviewed.

4.4 SSH service check

```
systemctl status ssh
```

Expected output: **active (running)** and **enabled**, with the log lines showing the service listening on port 22.

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5. Troubleshooting Guide

5.1 apt repository errors

Symptom: apt update or apt upgrade fails with errors such as 403 Forbidden or Malformed entry ... (URI).

Cause: A regional mirror may be temporarily out of sync, or the repository file contains a wrong URI.

Solution: Retry the command once. If the error persists, open the repository file and point the URIs to the main archive:

```
sudo nano /etc/apt/sources.list.d/ubuntu.sources
```

Make sure each block uses a valid URI, for example http://archive.ubuntu.com/ubuntu or http://security.ubuntu.com/ubuntu, then save (Ctrl+O, Enter) and exit (Ctrl+X). Run sudo apt update to verify the fix.

5.2 Interrupted package operations

Symptom: After a power failure or a forced restart, apt reports broken packages or leaves the system in an inconsistent state.

Solution: Repair the installation in this order:

```
sudo dpkg --configure -a
sudo apt --fix-broken install -y
sudo apt upgrade -y
```

5.3 SSH service not found

Symptom: systemctl status ssh returns Unit ssh.service could not be found.

Cause: The OpenSSH server package is not installed, usually because the option was skipped during installation.

Solution: Install the package and start the service as described in section 3.2.

5.4 Unresponsive virtual machine

Symptom: The virtual machine window stops accepting clicks or keyboard input.

Solution: Press the host key (Right Ctrl by default) to release the mouse, then click inside the window again. If the machine is truly frozen, use **Machine, Reset** from the VirtualBox menu, and repair any interrupted package operations afterward (section 5.2).

6. Best Practices

- Always use the latest LTS release of Ubuntu Server for new deployments, since it receives five years of security updates.
- Record the IP address, hostname, and account details during installation and keep them in a secure inventory document.
- Change the default passwords and never reuse passwords across systems.
- Enable OpenSSH during installation when remote administration is planned, and disable password login after SSH keys are configured.
- Apply updates immediately after installation and on a fixed schedule afterward.
- Take a snapshot of the virtual machine right after a clean installation, before making any major changes.
- Document every configuration change so that the next administrator can reproduce the environment.
- Detach the installation ISO after the first boot to prevent accidental reinstallation.

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7. References

- Ubuntu Server Documentation, https://ubuntu.com/server/docs
- Ubuntu Installation Guide, https://help.ubuntu.com/community/Installation
- Oracle VirtualBox User Manual, https://www.virtualbox.org/manual
- OpenSSH Manual Pages, https://www.openssh.com/manual.html
- Ubuntu Community Help, APT, https://help.ubuntu.com/community/AptGet